Narsi Reddy Sanikommu

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Seeking a mechanical design and hardware engineering full-time position.

Education

Johns Hopkins University | *M.S. Mechanical Engineering* — Baltimore, MD | Graduated 2026
NIT Trichy | *B.Tech. Mechanical Engineering* — Tiruchirappalli, India | Graduated 2020

Work Experience

BD Life sciences | *Mechanical and Hardware engineering COOP Intern* — Jul 2025 – Jan 2026

- Designed using **SolidWorks** and **GD&T** and validated using **Shimadzu**, a shuttle fixture for the BD COR GX shuttle, enabling **20,000-cycle** reliability testing vs. **500 cycles** previously, significantly improving test efficiency and reliability.
- Built a 3D sensor response contour model for the BD COR MX liquid waste detection sensor using **NumPy & SciPy**, achieving **95% prediction accuracy** of trigger volume and supporting design optimization decisions.
- Performed feasibility study and design iteration of the BD COR PX load cell using **SolidWorks & Flow Simulation**, enabling real-time measurement of sample, diluent, and reagent weights for improved error detection and spill prevention.
- Executed **RCA** on BD COR MX test fixture; replaced 3D-printed components with laser-cut acrylic and Designed a snap-fit reinforcement using **SolidWorks simulation** in **DLS Flex 80A**, eliminating warping without triggering tool revalidation.
- Designed and prototyped a magnetic brake fixture in **SolidWorks** and in-house **CNC** machining, to test motor torque and loading across all stepper motors in **BD COR** systems.

Becton Dickinson and company | *Design Engineering 2* — Jul 2020 – Aug 2023

- Led Swiss AR label changes, updating redline labels, releasing ECOs, and coordinating labeling across suppliers and plants.
- Identified 30 PFOA-impacted components among 8,000+ SKUs using **SAP, VBA, and custom macros** in 2 months.
- Led R&D effort for Stopcock manifold material changes, creating DV protocols and fixtures, reducing test time by 2 weeks.
- Led supplier packaging changes for HDS IV-line SKUs, coordinating **DV, IQ/OQ/PQ, FAIs, and incoming inspection**.
- Led Borealis material changes for BD PhaSeal P55/P50/P28 through V&V testing, Minitab analysis, and SAP releases.
- Updated luer test methods per **ISO 80369-7** via existing MSA studies & **MSA by analysis**, eliminating full MSA.
- Identified resterilizable SKUs using sterilization and DV data, preventing 3 months of back order during **Sterilmilano issue**.
- Supported **CAPA** for HDS P55's RCA using **DFSS, Ishikawa diagrams and CT scans**, improving leak testing without redesign.
- Designed, prototyped, and tested NFC and Stopcock concepts via SolidWorks, SLA prints, and **Instron** testing in 1.5 weeks.
- Developed a rapid prototype for a new HDS product in **1 week** at a cost of **$100** using existing BD PIVC platform components, accelerating concept validation and cutting reduced NPD timeline by **6 months**.
- Contributed to innovation pipeline with **68 invention disclosures** and **14 US patent filings**, including US12502516B2, US12186518B2, US12188581B2, and US12440662B2.

Siemens COE, NIT Trichy | *CAD Designer Intern* — May 2019 – Jun 2019

- Designed a complete E-bus model in **NX for BHEL**, iterating the design to reduce weight by **15%**, including reverse engineered steering and axle components from .step & .stl files.
- Generated meshes in **HyperWorks** and performed static structural analysis using **Ansys, Abaqus, and Nastran**.
- Conducted a simplified **crash test simulation** to validate structural integrity of the final design in **Ansys**.

Projects

Personal Projects — Jan 2026 – Present

- Fabricated a 10 cm hydrogel-based flexible robot actuated by segmented flow control using a custom microfluidic pump.
- Developed a PyBullet digital twin using extensible rod theory to model equilibrium shape and end-effector position.
- Researched a single-channel microfluidic platform for multi-site drug delivery using segmented air–liquid flow and SLA clear resin prototypes.

DR Gracias lab | *Research Assistant* — oct 2024 – Nov 2025

- Fabricated a soft hydrogel microneedle drug device using soft lithography for targeted drug delivery, in 10 minutes with improved mold-release yield from 40% to 80% over 3 design iterations.
- Developed a Python-OpenCV pipeline for hydrogel roll unfolding analysis across 50+ videos, reducing 1 week of manual ImageJ processing to an automated workflow completed in 1 day.
- Fabricated multilayer SU-8/metal microelectrode array shells for neuromorphic computing applications using photolithography, thermal evaporation, and electroplating.